Abhijit Kashyap

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.48
BASE PU College	Class XII (Karnataka State Board)	2023	93.83%
Manasarowar Pushkarini Vidyashrama	Class X (CBSE)	2021	96.16%

INDUSTRY EXPERIENCE

SIEMENS | *Soft Robotics Intern*

- Designed, manufactured and tested mechanical components and PCBs for Universal Testing Machine (**UTM**) to obtain stress-strain curves of hyper-elastic materials, rated for a maximum load of **800N** and displacement resolution of **0.7** microns
- Calibrated the UTM for structural elasticity using finite-element and experimental methods, limiting displacement measurement uncertainty to **±5 µm** (0.01%) under peak loading conditions
- Used stress-strain data of TPU as input for a non-linear finite element method based solver to model hyper-elastic materials
- Developed a print-in-place multi-material bearing to **reduce friction** and **wear** in tendons in soft robotic actuators
- Designed a test setup to measure tendon friction and wear during oscillatory motion over moving capstan for $\mathbf{10^6}$ cycles

PROJECT P-SUIT | *Military-grade pressure suit*

- Evaluated 4 materials for **suit exterior** based on water-resistance, air tightness, biocompatibility and cost
- Designed the **gas-exchange architecture** and pressure relief mechanism using one-way valves and an inlet pump
- Developed an **STM32**-based real-time **vital monitoring system**, integrating SpO2, heart-rate and respiration-rate sensors

PROJECTS

QuTTA: The Quadruped Bot | *Inter-IIT Tech Meet 13.0, IIT Bombay, December 2024*

- Represented Robotix IIT Hyderabad at Inter-IIT Tech Expo held at IIT Bombay, December 2024
- Built a servo-actuated 12-DOF **quadruped** bot imitating crawling and trotting gaits using inverse kinematics
- Redesigned the system architecture, reducing cost by **4** times, and projected development time by **2.5** times.
- Led electronics design and integration: custom power-distribution board design, actuator control and system debugging
- Fine-tuned a closed-loop **PID** controller for dynamic pitch and roll stabilization using 6-axis IMU feedback

Formula Student Race Car | *Aeolus Racing IIT Hyderabad*

- Represented IITH at **SUPRA SAE 2025**, a national-level Formula Student competition
- Led the **electrical domain**, overseeing vehicle electrical integration including ECU, sensors, ignition, pump and radiator circuits, E-Stops, brake light and fail-safe safety systems
- Contributed to the **cost report** team, preparing and presenting the business pitch and detailed cost breakdown of the car

Autonomous Rover for Warehouse Inventory Management | *Inter-IIT Tech Meet 14.0, IIT Patna, December 2025*

- Led the IITH team for the Inter-IIT Tech Meet 14.0 Mid-Prep Problem Statement by Eternal Limited
- Developed a **ROS2**-powered autonomous rover implementing **SLAM** in **Gazebo** for path planning and rack/obstacle detection
- Led **mechanical design** of a two-stage telescopic lead-screw mechanism for camera height control, reducing cost by **3.6** times
- Designed **electronics** and **PCBs** for locomotion, camera elevation and battery management subsystems

Thumb-Fighting Robot | *Under Dr Safvan Palathingal, Assistant Professor, Mechanical Engineering, IIT Hyderabad*

- Modelled a 2-DOF human thumb mechanism in **SolidWorks**, reducing 3D-printing filament requirement by **20%**
- Developed a tendon-driven **actuation mechanism** inspired by the tension-based operation of human thumb muscles
- Designed and fabricated the prototype to evaluate mechanical motion and tendon-based actuation

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, MATLAB
- **CAD:** SolidWorks, Siemens NX, Fusion 360, Solid Edge, Onshape
- **Electronics and FPGA:** KiCAD, TINA-TI, Vivado
- **Embedded Systems:** Arduino (Uno, Nano and Mega), ESP32, STM32, Raspberry Pi Pico, Raspberry Pi 5
- **Simulation and Analysis Tools:** Abaqus, TINA-TI, Vivado, Hexagon ADAMS
- **Robotics & Simulation:** ROS2, Gazebo, Abaqus, Hexagon ADAMS
- **3D Printing:** Experience with Creality Ender 3, Bambu Lab printers (A1, P1S, X1C, H2C), AnyCubic Kobra 3
- **Manufacturing:** Hands-on experience with the lathe, drilling press, angle grinder, jigsaw, and in arc welding

POSITIONS OF RESPONSIBILITY

- **Overall Head**, Robotix IIT Hyderabad
- **Electrical Domain Head**, Aeolus Racing
- **Mechatronics Core**, Tinkerers' Lab IIT Hyderabad
- **Core Member**, Aero IIT Hyderabad
- **Core Member**, Infocus IIT Hyderabad
- **Core in Writing Club**, Litsoc IIT Hyderabad
- **Class Representative**, B.Tech Biomedical Engineering

EXTRACURRICULAR

- **Chess:** Rated 1400 elo
- **Speedcubing:** 3x3 Rubik's Cube: PB single 7.1 s, average of five 11.2 s
- **Photography:** Core member in the photography club for 2 years